Summary of the 2nd International Flaky Test Workshop (FTW 2025)

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ABSTRACT

Test flakiness refers to the unpredictable behavior of software tests, where a test may pass or fail even when neither the test code nor the system under test has changed. This results in reduced developer productivity, costly and repeated test reruns, hidden or overlooked defects, and diminished confidence in test suites.

While the problem has received increasing attention from both practitioners and researchers in recent years, opportunities for focused discussion remain limited.

The 2nd International Flaky Test Workshop (FTW 2025) was held on April 27th, 2025, in conjunction with the 47th International Conference on Software Engineering (ICSE 2025). The program emphasized two main themes, *flakiness in specific and previously neglected domains* and *improving flakiness mitigation techniques*, and also included demonstrations of new datasets and tools to support research and practice.

1. INTRODUCTION

Test flakiness remains a persistent challenge, particularly in large-scale software projects that rely heavily on automated testing and continuous integration (CI). Its impact is especially pronounced in industrial contexts, where the reliability of test suites directly affects developer productivity and software quality.

The 1st International Flaky Test Workshop (FTW 2024) successfully established a dedicated venue for addressing this problem, with the aim of fostering dialogue between academia and industry. This exchange proved helpful in maintaining the relevance of ongoing research to practical needs while benefiting from empirical and methodological advances.

Building on that foundation, the 2nd International Flaky Test Workshop (FTW 2025) strengthened this collaboration. While test flakiness has attracted significant attention in recent years, with numerous empirical studies and tool prototypes, substantial challenges remain. FTW 2025 provided a timely forum to consolidate progress, broaden perspectives to previously neglected domains, and refine mitigation techniques, while also showcasing datasets and tools that could accelerate future research and practice.

The workshop was organized by Martin Gruber, a software engineer at BMW who recently completed his doctoral dissertation on test flakiness; Wing Lam, an assistant professor at George Mason University whose research focuses on characterizing and mitigating software bugs, including extensive work on flaky tests; Owain Parry, an early-career researcher with significant contributions to flaky test research, ranging from systematic literature reviews to machine learning–based detection techniques; and August Shi, an assistant professor at the University of Texas at Austin with a strong publication record on detecting, debugging, and repairing flaky tests. Collectively, the committee brought a blend of industrial experience and academic expertise, ensuring a program that was both rigorous and practically relevant.

2. KEYNOTE

The keynote, *Flaky Tests: Good Research with Great People*, was delivered by Michael Hilton, who summarized over a decade of research on flaky tests and reflected on his positive experiences collaborating with other researchers in the field. In 2021, Hilton co-authored *FlakeFlagger: Predicting Flakiness Without Rerunning Tests*, presented at ICSE 2021, introducing a tool to predict test flakiness without repeated executions. In 2022, he contributed to *Surveying the Developer Experience of Flaky Tests*, presented at ICSE 2022, which explored developers' challenges with flaky tests. Hilton has also been particularly active in studying flakiness in continuous integration environments. His talk highlighted key insights from these studies and emphasized practical approaches to understanding, detecting, and mitigating flaky tests.

3. PAPERS

We split the paper presentations into two sessions of three. The morning session, *Flakiness in Specific/Previously Neglected Domains*, focused on understanding flakiness in diverse contexts, including Rust projects, autonomous driving systems, and broader perspectives on test suite health. The afternoon session, *Improving Flakiness Mitigation Techniques*, examined strategies for detecting, prioritizing, and reducing flaky tests, covering topics such as the application of NonDex to modern Java projects, the perception and adoption of flakiness scores, and methods to reduce test re-runs by targeting order-dependent flaky tests.

3.1 Flakiness in Specific/Previously Neglected Domains

The first paper in this session was *A Preliminary Study of Fixed Flaky Tests in Rust Projects on GitHub* by Tom Schroeder, Minh Phan, and Yang Chen. This preliminary study investigates fixed flaky tests in Rust projects on GitHub, a language not extensively studied for flakiness. By inspecting 53 identified issues, the researchers found that asynchronous wait (33.9%), concurrency issues (24.5%), logic errors (9.4%), and network-related problems

(9.4%) were the predominant root causes of flakiness. Fixes often involved modifying the main code for common causes like async waits and concurrency, though some only reduced the likelihood of failure rather than completely removing flakiness.

The second paper was *Beyond Test Flakiness: A Manifesto for a Holistic Approach to Test Suite Health* by Phil McMinn, Muhammad Firhard Roslan, and Gregory Kapfhammer. This paper argues for a holistic approach to test suite health, asserting that flaky tests are just one of several indicators of an "unhealthy" test suite. It identifies nine key health indicators, including low code coverage, high pseudo-testedness, low mutation score, and high brittleness, noting that trade-offs and synergies exist among them. The authors advocate for further research into measuring and combining these indicators to provide actionable recommendations and automated tooling for improving test suite health.

The third paper was *Empirically Evaluating Flaky Tests for Autonomous Driving Systems in Simulated Environments* by Olek Osikowicz, Phil McMinn, and Donghwan Shin. This paper empirically investigates flaky tests in Autonomous Driving Systems (ADS) testing within simulated environments, comparing CARLA and MetaDrive. The results show that 31.3% of benchmark test scenarios in CARLA are potentially flaky due to non-deterministic simulations, particularly affecting collision-related safety requirements, while MetaDrive yielded no flaky tests. The paper discusses potential causes like asynchronous behavior and game engine reliance, implications for critical scenario generation, and proposes a four-step mitigation strategy.

3.2 Improving Flakiness Mitigation Techniques

The first paper in this session was *Evaluating NonDex for Modern Java Ecosystem* by Kaiyao Ke and Darko Marinov. This paper evaluates the updated NonDex tool, designed to detect Implementation-Dependent (ID) flaky tests that stem from incorrect assumptions about non-deterministic Java API specifications. The improved NonDex tool now supports modern Java versions and integrates with both Maven and Gradle build systems. An extensive evaluation on modern Java projects revealed a significant increase in the prevalence of flaky tests, with a case study demonstrating how flakiness can propagate through project dependencies, leading to a fix in Gradle's core code.

The second paper was *On the Perception and Adoption of Flakiness Scores* by Joanna Kisaakye, Mutlu Beyazıt, and Serge Demeyer. This work focuses on the perception and adoption of test flakiness scores, extending existing definitions of "flip rate" to "transition rate" to accommodate a greater variety of test outcomes (Pass, Fail, Error, Skip) at the system test level. The authors aim to investigate practitioner perception and adoption of these extended flakiness scores across different DevOps roles. This is part of a work-in-progress study with industrial partners to bridge the gap between research and practice.

The third paper was *Reduction of Test Re-runs by Prioritizing Potential Order Dependent Flaky Tests* by Hasnain Iqbal, Zerina Begum, and Kazi Sakib. This paper proposes a method to prioritize potential Order-Dependent (OD) flaky tests to reduce the computational cost associated with their detection. The approach focuses on analyzing shared static fields in test classes, which are a primary cause of OD flakiness, to identify and prioritize contributing test methods. In an experiment across 27 projects, this method successfully prioritized all OD tests in 23 projects, leading to an average reduction of 65.9% in test cases and 72.2% in re-runs.

4. DATASET/TOOL DEMONSTRATIONS

The workshop concluded with a session of demonstrations highlighting datasets and tools relevant to flaky tests. Owain Parry began with a brief overview of several existing flaky test datasets, followed by Joanna Kisaakye presenting her dataset. Kaiyao Ke then demonstrated the NonDex tool, and Wing Lam together with August Shi presented the IDoFT dataset. The session concluded with a discussion among all attendees on the requirements for high-quality open-source datasets to support future research in the field.

5. CONCLUSION

FTW 2025 built on the success of the inaugural workshop to provide a dedicated venue for discussing flaky tests, bringing together researchers and practitioners to share insights, experiences, and solutions. The keynote, paper sessions, and dataset/tool demonstrations highlighted advances in understanding flakiness, addressing neglected domains, and improving mitigation techniques.

We are grateful to the keynote speaker, the paper authors, and the program committee for their contributions, and we thank all attendees for their active participation. We also acknowledge the support of the ICSE 2025 organizing committee in making this workshop possible. We look forward to continuing this forum in future editions of FTW.